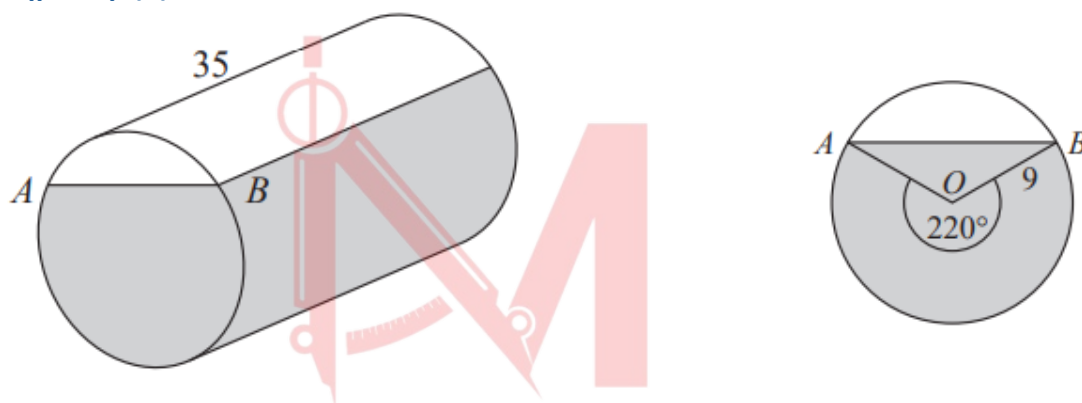


Mensuration

Worksheet (4024 – Paper 2)

Prisms and Uniform Solids

1. S09/qp02/q7(b)



A solid cylinder, made from a different type of wood, floats in water. The shaded region represents part of the surface of the cylinder that is in contact with the water. The right hand diagram shows the circular cross-section of one end. The centre of the circle is O and the water level reaches the points A and B on the circumference. Reflex angle $\widehat{AOB} = 220^\circ$. The cylinder has radius 9cm and length 35cm. Calculate

- i. the area of the curved surface of the cylinder that is in contact with the water, [2]
- ii. the surface area of one end of the cylinder that is in contact with the water, [4]
- iii. the distance between the water level AB and the top of the cylinder. [2]

2. S10/qp21/q9

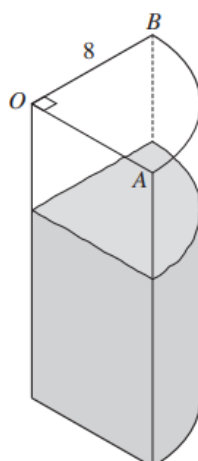


Diagram I

Diagram I shows a vessel in the shape of a prism. The cross-section OAB is a sector of a circle of radius 8cm and $\widehat{AOB} = 90^\circ$.

- a.
 - i. Calculate the perimeter of the sector OAB . [3]
 - ii. The vessel, which stands on a horizontal table, contains 800cm^3 of water, shown shaded in the diagram. Calculate the depth of the water in the vessel. [3]
- b.

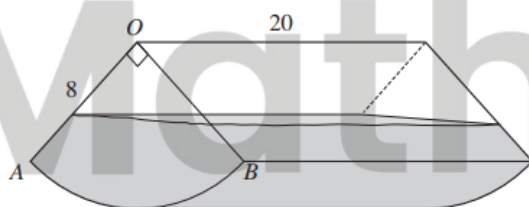


Diagram II

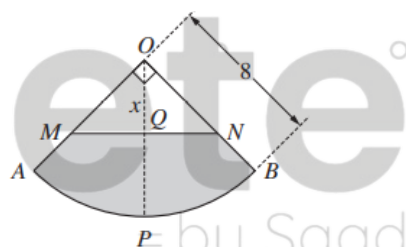


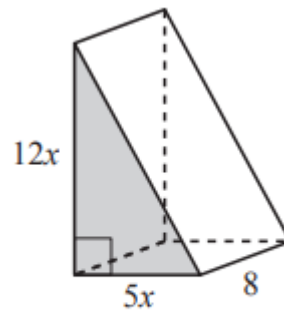
Diagram III

The vessel is now placed so that its curved surface is in contact with the horizontal table as shown in Diagram II. Diagram III shows the cross-section of the vessel with the shaded section representing the water. P is the midpoint of the arc AB and is in contact with the table. Q is the point on the water surface, MN , which is vertically above P . It is given that $OQ = x$ centimetres.

- i.
 - a. Write down an expression, in terms of x , for MN . [1]
 - b. Show that the shaded area in Diagram III is $(16\pi - x^2)$ square centimetres. [2]
- ii. The vessel still contains 800cm^3 of water. Given that the length of the vessel is 20cm , find the value of x . [3]

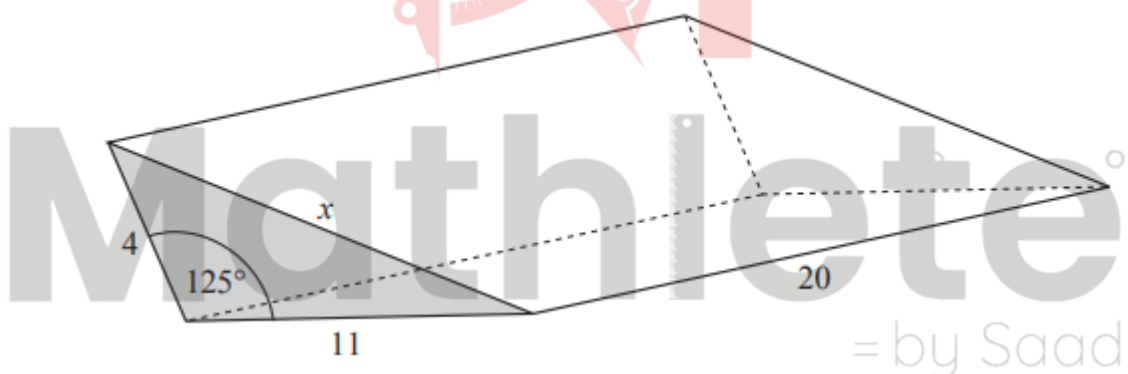
3. S13/qp21/q9(b)

Shape II is a prism of length 8cm with a triangular cross-section, shown shaded. Two sides of the shaded triangle are at right angles to each other and have lengths $5x \text{ cm}$ and $12x \text{ cm}$. Given that Shape II also has a volume of 1500cm^3 , find x . [2]



Shape II

4. S13/qp22/q12(b)



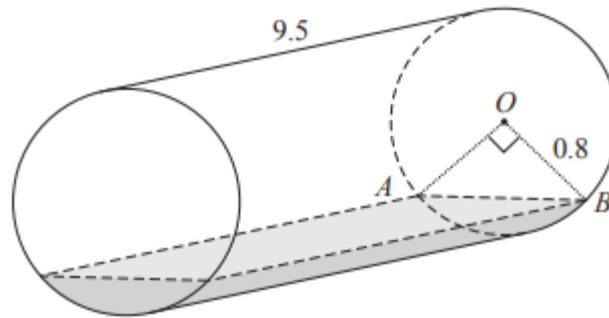
A triangular prism has length 20cm. The sides of the shaded cross-section are 4cm, 11cm and $x \text{ cm}$. The angle between the sides of length 4cm and 11cm is 125° .

- Calculate the area of the shaded cross-section. [2]
- Calculate the volume of the prism. [1]
- Calculate x . [4]
- Calculate the surface area of the prism. [2]

5. W13/qp21/q10(a)

A fuel tanker delivers fuel in a cylindrical container of length 9.5 m and radius 0.85 m.

- a. After several deliveries, the fuel remaining in the container is shown in the diagram.

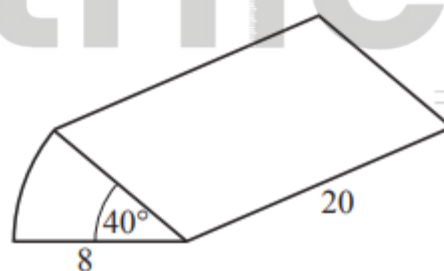


AB is horizontal, O is the centre of the circular cross-section and

$$\angle AOB = 90^\circ.$$

- Calculate the curved surface area of the container that is in contact with the fuel. [2]
- Calculate the volume of fuel remaining in the container. [4]
- Calculate this volume remaining as a percentage of the volume of the whole container. [3]

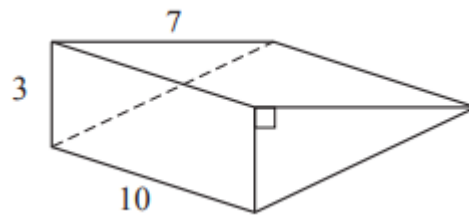
6. W14/qp21/q10(b)



The cross-section of a prism is a sector of a circle, radius 8 cm and angle 40° . The prism is 20 cm long. Calculate

- the area of the cross-section, [2]
- the total surface area of the prism. [4]

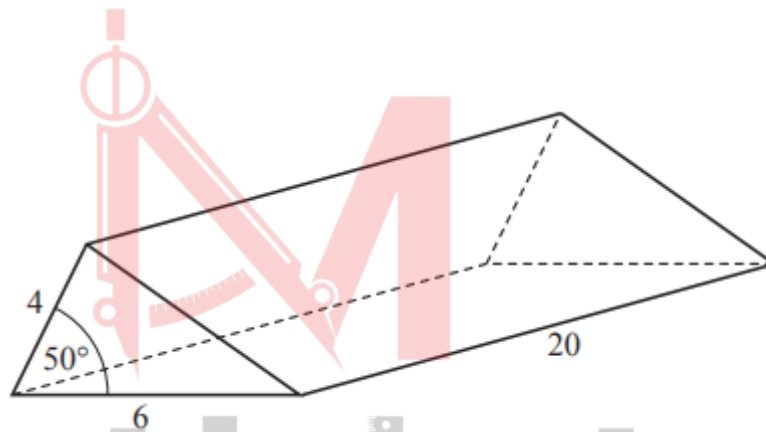
7. W14/qp22/q4(a)



The diagram shows a solid triangular prism. The dimensions are in metres.

- i. Calculate the volume of the prism. [2]
- ii. Calculate the total surface area of the prism. [4]

8. W15/qp21/q11(a)

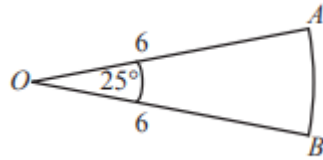


The diagram shows a solid triangular prism.
All lengths are given in centimeters.

- i. Calculate the area of the cross-section of the prism. [2]
- ii. Calculate the volume of the prism. [1]
- iii. Calculate the total surface area of the prism. [3]

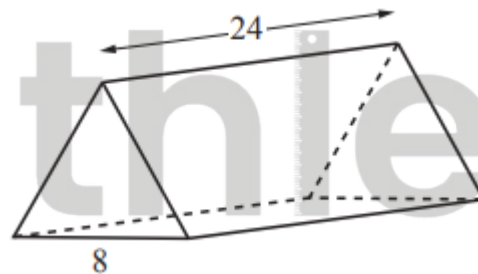
9. W15/qp22/q8(a)(b)

- a. OAB is a sector of a circle, center O , radius 66 cm.
 $\angle AOB = 25^\circ$



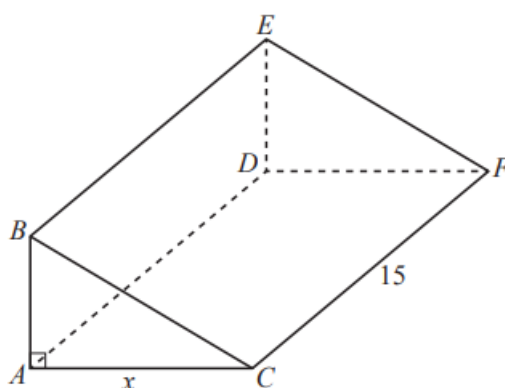
- i. Calculate the length of the arc AB . [2]
 - ii. Calculate the area of the sector OAB . [2]
- b. The sector OAB from part (a) is the cross-section of a slice of cheese.
The slice has a height of 55 cm.
- i. Calculate the volume of this slice of cheese. [1]
 - ii. Calculate the total surface area of this slice of cheese. [2]
 - iii. Another 25° slice of cheese has 3 times the height and twice the radius.
Calculate its volume. [2]

10. S16/qp22/q11(b)



The diagram shows a triangular prism of length 24 cm. Its cross-section is an equilateral triangle with sides 8 cm. Calculate the total surface area of the prism. [4]

11. W18/qp21/q10



The diagram shows a prism with a rectangular base of length 15cm and width x cm. The cross section of the prism is a right-angled triangle. The height of the prism is 4 cm less than its width. The volume of the prism is 440 cm^3 .

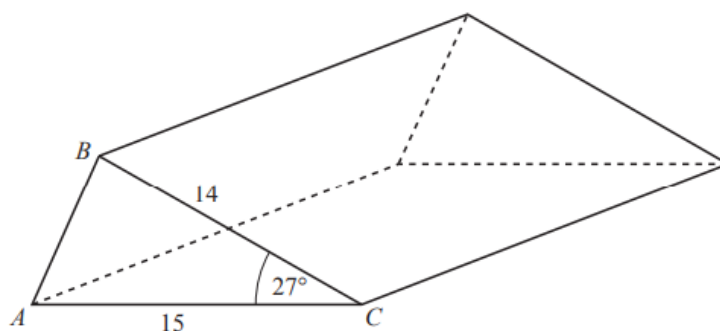
(a) Show that $3x^2 - 12x - 176 = 0$. [3]

(b) Solve the equation $3x^2 - 12x - 176 = 0$. Show your working and give your answers correct to 2 decimal places. [3]

(c) Find the height of the prism. [1]

(d) A line is drawn on the surface of the prism from B to F . Another line is drawn on the surface of the prism from A to F . Calculate angle AFB . [4]

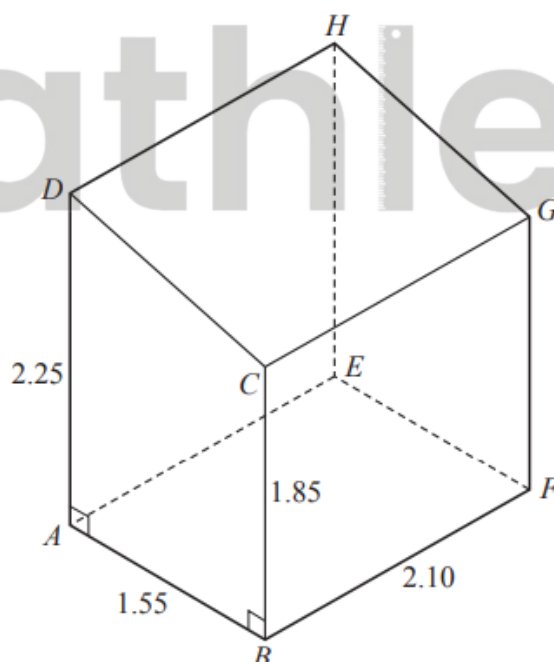
12. S19/qp22/q9



The diagram shows a triangular prism. $AC = 15\text{ cm}$, $BC = 14\text{ cm}$ and angle $ACB = 27^\circ$.

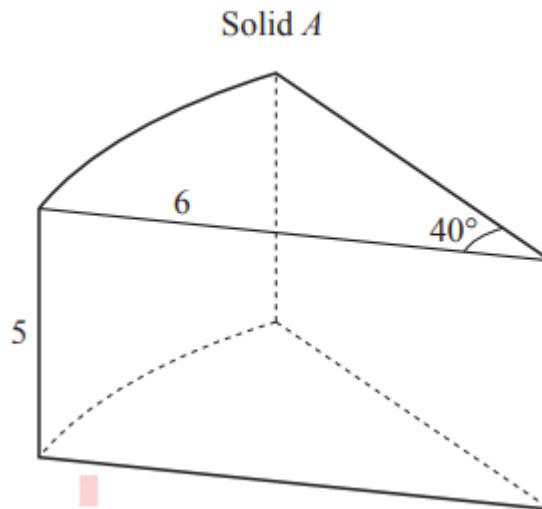
- Calculate AB . [3]
- The length of the prism is $p\text{ cm}$ and the volume of the prism is 1000 cm^3 . Calculate p . [3]
- The prism is to be packed in a carton. The carton is a cuboid of size 15 cm by $p\text{ cm}$ by $h\text{ cm}$. Calculate the smallest possible value of h . [2]

13. W20/qp21/q5(a)



The diagram shows a garden shed positioned on horizontal ground. It is in the shape of a prism with trapezium $ABCD$ as its cross-section. The base of the shed, $ABFE$, is a rectangle. $AB = 1.55\text{ m}$, $AD = 2.25\text{ m}$, $BC = 1.85\text{ m}$ and $BF = 2.10\text{ m}$. Calculate the volume of the shed. [3]

14. W20/qp21/q7(b)



The cross-section of solid A is the sector of a circle of radius 6cm and angle 40° . The height of solid A is 5 cm.

- i. Calculate the total surface area of solid A . [4]
- ii. Solid B is mathematically similar to solid A .
The ratio volume of solid A : volume of solid B = $27 : 1$. Calculate the surface area of solid B . [2]

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Answer Key

1. S09/qp02/q7(b)

	(b)	(i)	$\frac{220}{360} \times 2\pi \times 9 \times 35$ 1208 to 1210	M1	2	POSSIBLE GRAD ANSWERS (ii) 188 to 188.3 from $\frac{1}{2} \times 9^2 \times \sin 140 (= 32.7 \dots)$; 177 to 178 from $81 \times \sin 70 \times \sin 20 (= 22.3 \dots)$ (iii) 4.9 from $\cos 70$; 6.2 from $\sin 20$ sc1 for 4.08 ... or for 2.7 ... - If A0, then sc1 for 155 to 156 seen or for 25.9 to 26.1 seen } }sc1 for 3.07 to 3.08 seen
		(ii)	$\frac{220}{360} \times \pi \times 9^2 (= 155.50 \dots)$ $\frac{1}{2} \times 9^2 \times \sin 140 (= 26.03 \dots)$ 181 to 182	M1 M1 A2		
		(iii)	$d = 9 - 9 \cos 70$ $= 5.92$ to 5.93	M1 A1	2	
					[12]	

2. S10/qp21/q9

9	(a)	(i)	$\frac{90}{360} \times \pi \times 16$ +16 28.56 to 28.6(0) cm	M1	Correct formula and 90° used
				M1 A1	Indep. Attempt to add $2 \times$ radius
		(ii)	$\frac{90}{360} \times \pi \times 8^2$ [Their $\frac{90}{360} \times \pi \times 8^2] \times h$ $= 800$ soi $h = 15.9(0)$ to 15.92 cm	M1	Area of cross-section
				M1 A1	Indep. Forms equation = by Saad
	(b)	(i)	(a) $MN = 2x$	B1	Expect justification and a subtraction
			(b) Area of triangle = $\frac{1}{2}$ their $(2x \times x)$ Area of sector = 16π and Subtraction	M1 A1	
		(ii)	$20(16\pi - x^2) = 800$ $x^2 = 10.2\dots$ to 10.3 $x = 3.2(0)$ to 3.21 cm	M1 A1 A1	Forms equation Correct method of solution
				[12]	

3. S13/qp21/q9(b)

(b)	$x = 2.5$	2	M1 for $\frac{1}{2} \times 12x \times 5x$ or better for cross section
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4. S13/qp22/q12(b)

(b) (i)	18(.0) to 18.03 cm ²	2	M1 for $\frac{1}{2} \times 4 \times 11 \times \sin 125$
(ii)	360 to 360.6 cm ³	1ft	
(iii)	$x = 13.69$ to 13.7	4	M1 for $4^2 + 11^2 \pm (2) \times 4 \times 11 \times \cos 125$ M1 for $x^2 = 4^2 + 11^2 - 2 \times 4 \times 11 \times \cos 125$ or better A1 for 187.4 – 187.5
(iv)	609.8 to 610.1 cm ²	2	M1 for at least 4 correct areas

5. W13/qp21/q10(a)

10	(a) (i) 11.9	2	B1 for $k \times 2\pi r \times h$
	(ii) 1.73 or 1.74	4	M1 for $\frac{1}{2} \times 0.8 \times 0.8 (\times \sin 90)$ oe and M1 for $(\frac{90}{360})\pi \times 0.8^2$ and M1 for (their 0.5026 – their 0.32) $\times 9.5$
	(iii) 9.1% ft	2ft	M1 for $\frac{(a)(ii)}{19.1} \times 100$

6. W14/qp21/q10(b)

(b) (i)	22.3	2	M1 for $\frac{40}{360} \times \pi \times 8^2$
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7. W14/qp22/q4(a)

4	(a) (i) 105	2	B1 for $\left(\frac{1}{2}\right) \times 7 \times 3 \times 10$ or M1 for Area of cross section $\times 10$ soi
	(ii) 197.2 (m ²)	4	M1 for $3^2 + 7^2$ and M1 for area of one triangular face and M1 for area of one rectangular face

8. W15/qp21/q11(a)

11 (a) (i)	9.19[...]	2	M1 for $\frac{1}{2} \times 4 \times 6 \times \sin 50$
(ii)	183 to 184	1ft	ft $20 \times \text{their } 9.19$
(iii)	310 to 310.5	5ft	ft $292 + 2 \times \text{their } 9.19$ B3 for 4.60 or 4.59[8...] or M2 for $4^2 + 6^2 - 2 \times 4 \times 6 \times \cos 50$ or M1 for cosine formula with one error AND M1 for $20 \times (4 + 6 + \text{their } 4.60) + 2 \times \text{their } 9.19$ oe

9. W15/qp22/q8(a)(b)

8 (a) (i)	2.62	2	M1 for $\frac{25}{360} \times 2\pi \times 6$
(ii)	7.85	2	M1 for $\frac{25}{360} \times \pi \times 6^2$
(b) (i)	39.3	1ft	
(ii)	88.8	3ft	B1 for 30 or 60 or M1 for $5 \times (a)(i)$ and indep M1 for $2 \times (a)(ii)$
(iii)	471 to 472	2ft	B1 for height = 15 and radius = 12 soi

10. S16/qp22/q11(b)

(b)	630 to 632	4	M1 for $\frac{1}{2} \times 8 \times 8 \times \sin 60$ or $\frac{1}{2} \times 8 \times \sqrt{48}$ oe M1 for 8×24 soi or 192 soi M1 for $3 \times 8 \times 24 + 2 \times \text{their}$ (triangle area)
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11. W18/qp21/q10

10(a)(i)	$\frac{1}{2}x(x-4) \times 15 = 440$	M2	B1 for height = $(x-4)$
10(a)(ii)	Correct expansion and simplification leading to $3x^2 - 12x - 176 = 0$	A1	
10(b)(i)	$\frac{-(-12) \pm \sqrt{(-12)^2 - 4 \times 3 \times -176}}{2 \times 3}$	B2	B1 for $\sqrt{(-12)^2 - 4 \times 3 \times -176}$ or for $\frac{-(-12) \pm \sqrt{\text{their } 2256}}{2 \times 3}$
10(b)(ii)	9.92 and -5.92	B1	
10(c)	5.92	1	FT (their positive root - 4) if result positive
10(d)	18.2 or 18.21 to 18.22...	4	M2 for $[AF =] \sqrt{15^2 + \text{their } 9.92^2}$ Or for $[BF =]$ $\sqrt{\text{their } 9.92^2 + \text{their } 5.92^2 + 15^2}$ oe or M1 for $AF^2 = 15^2 + \text{their } 9.92^2$ soi or for $BF^2 = \text{their } 9.92^2 + \text{their } 5.92^2 + 15^2$ oe AND M1 for $\tan [AFB] = \frac{\text{their } 5.92}{\text{their } 17.98}$ or $\sin [AFB] = \frac{\text{their } 5.92}{\text{their } 18.93}$

12. S19/qp22/q9

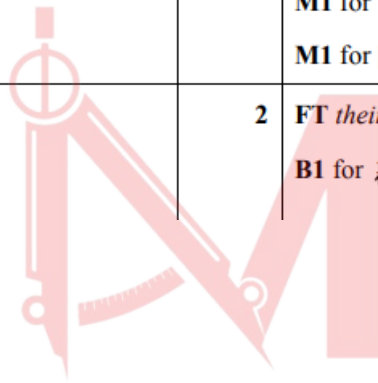
9(a)	6.84 or 6.839 to 6.84[0]	3	B2 for 46.7 to 46.8 or M1 for $15^2 + 14^2 - 2 \times 14 \times 15 \times \cos 27$
9(b)	21[.0] or 20.97 to 20.98	3	M2 for $\frac{1000}{\frac{1}{2} \times 14 \times 15 \times \sin 27}$ oe or M1 for $\frac{1}{2} \times 14 \times 15 \times \sin 27$ oe isw
9(c)	6.36 or 6.355 to 6.356	2	M1 for $\sin 27 = \frac{h}{14}$ oe

13. W20/qp21/q5(a)

5(a)	6.67 or 6.672 to 6.673	3	M2 for $\frac{1}{2} \times (2.25 + 1.85) \times 1.55 \times 2.1$ oe or M1 for $\frac{1}{2} \times (2.25 + 1.85) \times 1.55$ oe or $\frac{1}{2} \times 1.55 \times 2.1 \times (2.25 - 1.85)$ oe
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14. W20/qp21/q7(b)

7(b)(i)	106 to 106.1	4	M1 for $[2 \times] \frac{40}{360} \times \pi \times 6^2$ soi M1 for $\frac{40}{360} \times 2 \times \pi \times 6 \times 5$ soi M1 for $[2 \times] 5 \times 6$ soi
7(b)(ii)	11.7 to 11.8	2	FT <i>their</i> (b)(i) $\div 9$ B1 for $\sqrt[3]{\frac{1}{27}}$ or $\sqrt[3]{27}$ soi



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